PRACTICAL 1: ADVANCED LOG ANALYSIS

(SOC Log Correlation, Anomaly Detection & Enrichment)

Objective

The objective of this practical is to analyze security logs using SOC tools and techniques to:

- Correlate authentication logs with network activity
- Detect anomalies using rule-based conditions
- Enrich logs with contextual threat intelligence
- Understand real-world SOC log investigation workflow

Tools Used

The following tools were used as specified in the practical:

- Elastic Security for log ingestion, correlation, and rule creation
- Security Onion for log monitoring and analysis concepts
- Google Sheets for structured representation and reporting
- GeoIP lookup services for log enrichment

Implementation

Log Collection & Ingestion (Elastic Security)

- Logs were generated representing Windows authentication events and network traffic.
- These logs were ingested into **Elastic Security SIEM** for analysis.
- Fields included:
  - Event ID (4625, 4624, 61109)
  - Source IP
  - Destination IP
  - Bytes transferred

Log Correlation

Using Elastic Security dashboards, logs were filtered and correlated based on:

- Same Source IP (192.168.1.100)
- Sequence of events (Failed login → repeated attempts → outbound traffic)

Correlated Log Table:

Timestamp	Event ID	Source IP	Destination IP	Notes
11:58:21	4625	192.168.1.100	-	Failed login attempt
11:59:10	4625	192.168.1.100	-	Multiple failed login attempts
12:00:00	61109	192.168.1.100	8.8.8.8	Suspicious outbound traffic

Result:

The correlation indicates possible brute-force activity followed by suspicious network communication.

3.3 Anomaly Detection (Elastic Security Rule)

An anomaly detection rule was created in Elastic Security SIEM.
Rule Configuration:
Rule Name: High Data Transfer Detection
Condition: bytes_out > 1MB within 1 minute
Severity: High
Action: Generate Alert

Detected Event:

Source IP	Bytes Out	Status
192.168.1.100	1500000	ANOMALY

Result:
The system detected abnormal outbound traffic exceeding 1MB, indicating potential data exfiltration.

Log Enrichment (GeoIP Analysis)

Logs were enriched using GeoIP lookup to add geographical context to IP addresses.

IP Address	Country	City
8.8.8.8	USA	Mountain View

Result:
Enrichment helped identify the external destination location, improving investigation context.

Observations

- Multiple failed login attempts were observed from a single source IP
- This was followed by suspicious outbound network traffic
- An anomaly was detected due to unusually high data transfer
- GeoIP enrichment revealed external communication with a US-based IP

Conclusion

The practical successfully demonstrates real-world SOC operations using Elastic Security SIEM. Log correlation helped identify attack patterns, anomaly detection highlighted suspicious behavior, and enrichment provided contextual intelligence. This workflow is essential for effective threat detection and incident response in security operations centers.